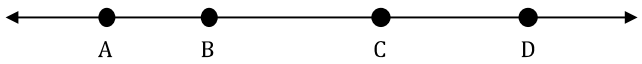
2.17 Geometry–Lines & Triangles

207. If each of the two lines l and m are parallel to line n, which of the following MUST be correct?

I. Lines l, m and n lie in the same plane.
II. Lines l and m are parallel to one another.
III. Line l is the same as line m.

(A) Only I
(B) Only II
(C) Only III
(D) Only I and II
(E) Only II and III

208. On the line segment AD shown below, AB $=\frac{1}{2}$CD and BD $=\frac{3}{2}$AC. If BC = 24, then AD =



(A) 24
(B) 48
(C) 72
(D) 96
(E) 120

209. R, S, T, and U are points on a line, and U is the midpoint of line segment ST. If the lengths of line segments RS, RT, and ST are 20, 4, and 16, respectively, what is the length of line segment RU?

(A) 6
(B) 8
(C) 12
(D) 14
(E) 16

210. A cash register in a certain clothing store is the same distance from two dressing rooms in the store. The distance between the two dressing rooms is 16 feet, which of the following could be the distance between the cash register and either dressing room?

I. 6 feet
II. 12 feet
III. 24 feet

(A) I only
(B) II only
(C) III only

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(D) I and II

(E) II and III

211. A certain right triangle has sides of length x, y and z, where $x < y < z$. If the area of this triangular region is 1, which of the following indicates all of the possible values of y?

(A) $y > \sqrt{2}$

(B) $\frac{\sqrt{3}}{2} < y < \sqrt{2}$

(C) $\frac{\sqrt{2}}{3} < y < \frac{\sqrt{3}}{2}$

(D) $\frac{\sqrt{3}}{4} < y < \frac{\sqrt{2}}{3}$

(E) $y < \frac{\sqrt{3}}{4}$

212. A certain right triangle has sides of length x, y and z, where $x < y < z$. If the area of this triangular region is 1, which of the following indicates all of the possible values of z?

(A) $z > 2$

(B) $\sqrt{2} < z < 2$

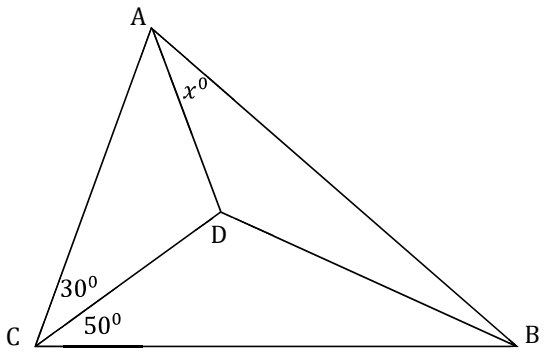
(C) $\sqrt{2} < z < \sqrt{3}$

(D) $1 < z < \sqrt{2}$

(E) $z < 1$

213. In the figure below, DA = DB = DC. What is the value of x?

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(A) 10

(B) 20

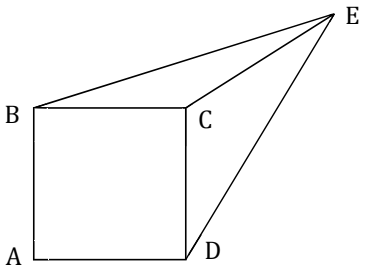
(C) 30

(D) 40

(E) 50

214. In the figure below, each side of square ABCD has length 1, the length of line segment CE is 1, and the length of line segment BE is equal to the length of line segment DE. What is the area of the triangular region BCE?

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(A) $\frac{1}{3}$

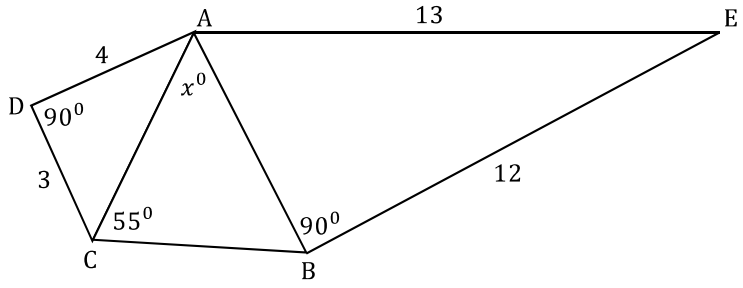
(B) $\frac{\sqrt{2}}{4}$

(C) $\frac{1}{2}$

(D) $\frac{\sqrt{2}}{2}$

(E) $\frac{3}{4}$

215. In the figure shown below, what is the value of x?



(A) 55

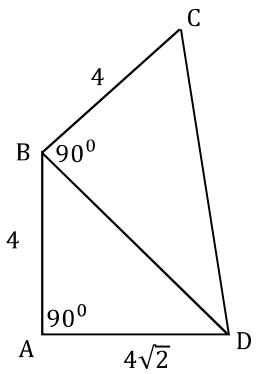
(B) 60

(C) 65

(D) 70

(E) 75

216. In the figure below, what is the perimeter of triangle BCD?



(A) $4 + 4\sqrt{3}$

(B) 12

(C) $12 + 4\sqrt{3}$

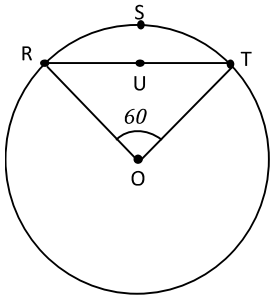
(D) $8 + 8\sqrt{3}$

(E) $16\sqrt{2}$

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2.18 Geometry–Circles

217. If the circle below has centre O and length of the arc RST is 18π, what is the perimeter of the region RSTU?



(A) $12\pi + 18$

(B) $12\pi + 27$

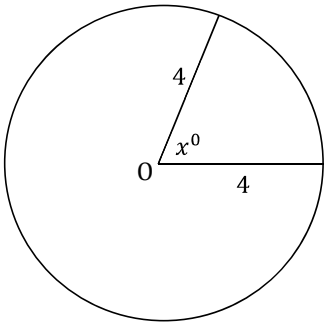
(C) $18\pi + 27$

(D) $18\pi + 54$

(E) $18\pi + 108$

218. In the figure below, O is the center of the circle**.** If the area of the sector containing the angle x^o is 2π, what is the value of x?

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(A) 22.5

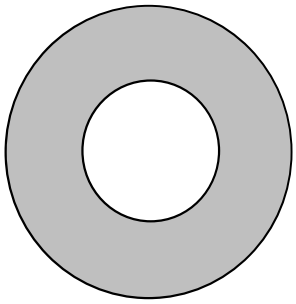
(B) 30.0

(C) 45.0

(D) 60.0

(E) 90.0

219. In the figure shown below, if the area of the shaded region is 3 times the area of the smaller circular region, then the circumference of the larger circle is how many times the circumference of the smaller circle?



(A) 4

(B) 3

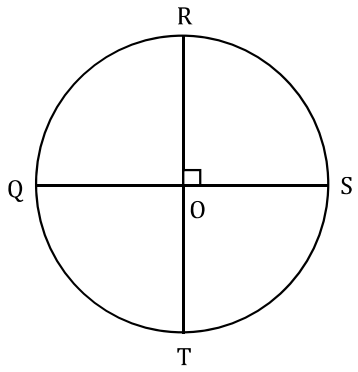
(C) 2

(D) $\sqrt{3}$

(E) $\sqrt{2}$

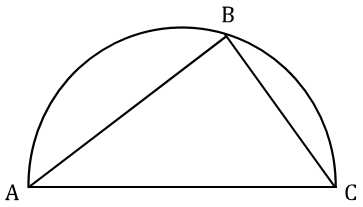
220. In the figure shown below, line segments QS and RT are diameters of the circle. If the distance between Q and R is $\frac{8}{\sqrt{2}}$, what is the area of the circle?

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(A) 4π
(B) 8π
(C) 16π
(D) 32π
(E) 64π

221. In the figure shown below, the triangle ABC is inscribed in a semicircle. If the length of line segment AB is 8 and the length of line segment BC is 6, what is the length of arc ABC?



(A) 15π
(B) 12π
(C) 10π
(D) 7π
(E) 5π

222. An equilateral triangle that has an area of $9\sqrt{3}$ is inscribed in a circle. What is the area of the circle?

(A) 6π
(B) 9π
(C) 12π
(D) $9\sqrt{3}\pi$
(E) $18\sqrt{3}\pi$

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2.19 Geometry–Polygon

223. A circular mat with diameter 20 inches is placed on a square tabletop, having its sides equal to 24 inches. Which of the following is closest to the fraction of the tabletop NOT covered by the mat?

(A) $\frac{1}{2}$

(B) $\frac{3}{5}$

(C) $\frac{2}{3}$

(D) $\frac{1}{4}$

(E) $\frac{9}{20}$

224. Rectangular floors having perimeter of 16 meters are to be covered with carpet squares that measure 1 meter by 1 meter each, costing \$6 apiece. What is the maximum possible cost for the number of carpet squares needed to cover any such rectangular floor if the sides of the floors are integers?

(A) \$42

(B) \$72

(C) \$90

(D) \$96

(E) \$120

225. A rectangular photograph is surrounded by a border that is 1 inch wide on each side. The total area of the photograph and the border is m square inches. If the border had been 2 inches wide on each side, the total area would have been $(m + 52)$ square inches. What is the perimeter of the photograph, in inches?

(A) 34

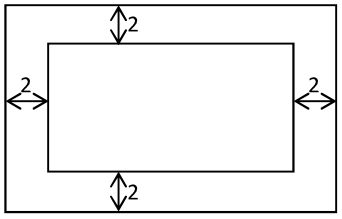
(B) 36

(C) 38

(D) 40

(E) 42

226. A rectangular picture is surrounded by a border, as shown in the figure below. Without the border, the length of the picture is twice its width. If the area of the border is 196 square inches, what is the length, in inches, of the picture, excluding the border?



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(A) 10

(B) 15

(C) 30

(D) 40

(E) 60

227. Two brothers inherited a rectangular field of dimension 80 feet by 120 feet. If they decide to split the land into two equal rectangles, then what is the minimum cost required to fence one such half at the rate of $2 per feet?

(A) $240

(B) $280

(C) $320

(D) $560

(E) $640

228. A rectangular park has a perimeter of 560 feet and a diagonal measurement of 200 feet. What is its area, in square feet?

(A) 19200

(B) 19600

(C) 20000

(D) 20400

(E) 20800

229. A solid yellow stripe is to be painted in the middle of a certain highway. If 1 gallon of paint covers an area of p square feet of highway, how many gallons of paint will be needed to paint a stripe t inches wide on a stretch of highway m miles long? (1 mile = 5,280 feet, and 1 foot = 12 inches)

(A) $\frac{5280mt}{12p}$

(B) $\frac{5280pt}{12m}$

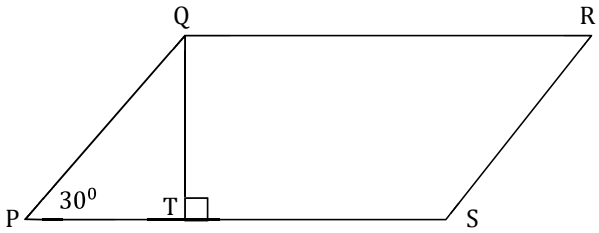
(C) $\frac{5280mpt}{12}$

(D) $\frac{5280 * 12m}{pt}$

(E) $\frac{5280 * 12p}{mt}$

230. In the parallelogram PQRS shown below, if PQ = 4 and QR = 6, what is the area of PQRS?

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(A) 8

(B) 12

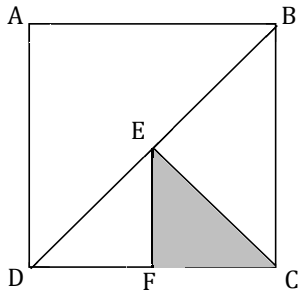
(C) 24

(D) $8\sqrt{3}$

(E) $12\sqrt{3}$

231.

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In square ABCD above, if DE = EB and DF = FC, then the area of the shaded region is what fraction of the area of square region ABCD?

(A) $\frac{1}{16}$

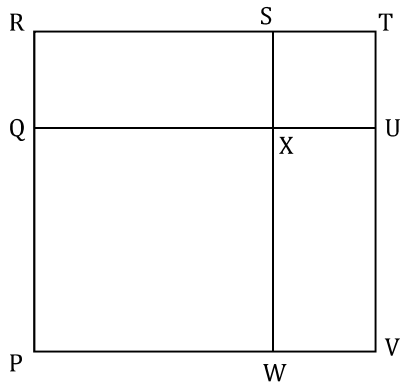
(B) $\frac{1}{8}$

(C) $\frac{1}{6}$

(D) $\frac{1}{4}$

(E) $\frac{1}{3}$

232. In the figure shown below, the area of square region PRTV is 81, and the ratio of the area of square region XSTU to the area of square region PQXW is 1 to 4. What is the length of segment RS?



(A) 5.0

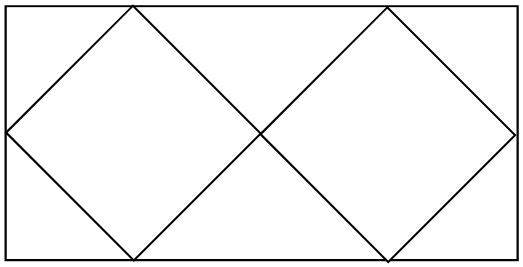
(B) 5.5

(C) 6.0

(D) 6.5

(E) 7.0

233.



In the figure above, two identical squares are inscribed in the rectangle. If the perimeter of the rectangle is $18\sqrt{2}$, then what is the perimeter of each square?

(A) $8\sqrt{2}$

(B) 12

(C) 16

(D) $12\sqrt{2}$

(E) 18

234. A certain number of desks and bookshelves, at least one each, are to be placed along a library wall that is 16 meters long. Each desk is 2 meters long, and each bookshelf is 1.5 meters long. If the maximum possible number of desks and bookshelves are to be placed along the wall, then the space along the wall that is left over will be how many meters long?

(A) 0.5

(B) 1.0

(C) 1.5

(D) 2.0

(E) 3.0

235. A thin piece of wire 40 meters long is cut into two pieces. One piece is used to form a circle with radius r, and the other is used to form a square. If no wire is left over, which of the following represents the total area, in square meters, of the circular and the square regions in terms of r?

(A) πr^2

(B) $\pi r^2 + 10$

(C) $\pi r^2 + \frac{1}{4}\pi^2 r^2$

(D) $\pi r^2 + (40 - 2\pi r)^2$

(E) $\pi r^2 + \left(10 - \frac{1}{2}\pi r\right)^2$

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2.20 Geometry–3 Dimensional

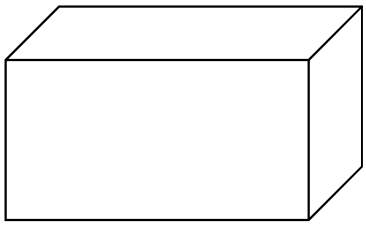
236. A certain right circular cylinder has a radius of 5 inches. A certain quantity of liquid fills this cylinder to a height of 9 inches. When all of this liquid is poured into a second right circular cylinder, the liquid fills the second cylinder to a height of 4 inches. What is the radius of the second cylinder, in inches?

(A) 6.0
(B) 6.5
(C) 7.0
(D) 7.5
(E) 8.0

237. A circular rim 28 inches in diameter rotates the same number of inches per second as a circular rim 35 inches in diameter. If the smaller rim makes x revolutions per second, how many revolutions per minute does the larger rim make in terms of x?

(A) $\frac{48\pi}{x}$
(B) $75x$
(C) $48x$
(D) $24x$
(E) $\frac{x}{75\pi}$

238. In the rectangular solid below, the three faces shown have areas 12, 15, and 20. What is the volume of the solid?



(A) 60
(B) 120
(C) 450
(D) 1800
(E) 3600

239. The interior of a rectangular carton is designed by a certain manufacturer to have a volume of x cubic feet and a ratio of length to width to height of 3 : 2 : 2. In terms of x, which of the following equals the height of the carton, in feet?

(A) $\sqrt[3]{x}$
(B) $\sqrt[3]{\frac{2}{3}x}$
(C) $\sqrt[3]{\frac{3}{2}x}$

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(D) $\frac{2}{3}\sqrt[3]{x}$

(E) $\frac{3}{2}\sqrt[3]{x}$

240. Two oil cans, X and Y, are right circular cylinders, and the height and the radius of Y are each twice those of X**.** If the oil in can X, which is filled to capacity, sells for $2, then at the same rate, how much does the oil in can Y sell for, if Y is filled to only half its capacity?

(A) $1

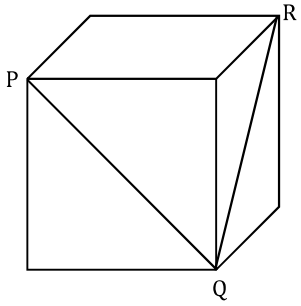
(B) $2

(C) $4

(D) $8

(E) $16

241. For the cube shown below, what is the degree measure of ∠PQR?



(A) 30°

(B) 45°

(C) 60°

(D) 75°

(E) 90°

242. A solid cube is placed in a cylindrical container. Which of the following percent values COULD possibly represent the ratio of the volume of the cylinder not occupied by the cube to the volume of the cylinder? (Assume the value of π to be 3)

(A) 16%

(B) 25%

(C) 28%

(D) 32%

(E) 36%

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2.21 Co-ordinate geometry

243. In the coordinate plane, a diameter of a circle has the end points $(-3, -6)$ and $(5, 0)$. What is the area of the circle?

(A) 5π
(B) $10\sqrt{2}\pi$
(C) 25π
(D) 50π
(E) 100π

244. A straight line in the XY-plane has a slope of 2 and a Y-intercept of 2. On this line, what is the X-coordinate of the point whose Y-coordinate is 500?

(A) 249
(B) 498
(C) 676
(D) 823
(E) 1002

245. In the XY-plane, a line n passes through the origin and has a slope 4. If points $(1, c)$ and $(d, 2)$ are on the line n, what is the value of $\frac{c}{d}$?

(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) 2
(D) 4
(E) 8

246. In the XY-plane, the point $(-2, -3)$ is the center of a circle. The point $(-2, 1)$ lies inside the circle and the point $(4, -3)$ lies outside the circle. If the radius of the circle is an integer, then what is the value of r?

(A) 6
(B) 5
(C) 4
(D) 3
(E) 2

247. In the XY-plane, the points (c, d), $(c, -d)$ and $(-c, -d)$ are three vertices of a certain square. If $c < 0$ and $d > 0$, which of the following points is in the same quadrant as the fourth vertex of the square?

(A) $(-5, -3)$
(B) $(-5, 3)$
(C) $(5, -3)$

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(D) $(3, -5)$

(E) $(3,\ 5)$

248. In the XY-plane, the vertices of a triangle have coordinates $(0,\ 0)$, $(3,\ 3)$ and $(7,\ 0)$. What is the perimeter of the triangle?

(A) $\sqrt{34}$

(B) $\sqrt{43}$

(C) 13

(D) $7 + 6\sqrt{2}$

(E) $12 + 3\sqrt{2}$

249. If the points $(a, 0)$, $(0, b)$ and $(1,\ 1)$ are collinear, what is the value of $\left(\frac{1}{a} + \frac{1}{b}\right)$?

(A) -1

(B) 0

(C) 1

(D) 2

(E) 3

250. In the XY-plane, what is the area of the triangle formed by the line $3y - 4x = 24$ and the X and Y axes?

(A) 6

(B) 14

(C) 24

(D) 36

(E) 48

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